Worldsheet Conventions for Type IIB String Theory

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1 Worldsheet CFT conventions

Euclidean worldsheet coordinates are denoted by z and $\bar{z}$. Target-space vector indices M, N take values in $\{0, 1, \ldots, 9\}$. We use the ten-dimensional Minkowski metric

$$\eta_{MN} := \mathrm{diag}(-1, +1, \ldots, +1). \tag{1.1}$$

We set

$$\alpha' = 1. \tag{1.2}$$

Some other general CFT_2 conventions we adopt are as follows:

1. Every coincident-point product in these notes is understood to be normal ordered by point splitting. Thus the notation $A(w)B(w)$ means

$$A(w)B(w) := \lim_{z\to w} \left(A(z)B(w) - \mathrm{Sing}_{z\to w}[A(z)B(w)] \right). \tag{1.3}$$

2. When a composite field is defined by an equation whose left-hand side is $C(z)$, every field on the right-hand side is evaluated at z unless a different argument is displayed explicitly.

3. If T is the stress tensor, a holomorphic operator $\mathcal{O}_h$ has conformal weight h when

$$T(z)\mathcal{O}_h(w) \sim \frac{h\,\mathcal{O}_h(w)}{(z-w)^2} + \frac{\partial\mathcal{O}_h(w)}{z-w}. \tag{1.4}$$

4. Let $T(z)$ be the stress tensor, let $T_F(z)$ be its supercurrent, and let c be the central charge. We define the normalization of these operators by the following OPEs

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}, \tag{1.5}$$

$$T(z)T_F(w) \sim \frac{3}{2}\,\frac{T_F(w)}{(z-w)^2} + \frac{\partial T_F(w)}{z-w}, \tag{1.6}$$

$$T_F(z)T_F(w) \sim \frac{c/6}{(z-w)^3} + \frac{T(w)}{2(z-w)}. \tag{1.7}$$

   Equations (1.5)–(1.7) match the convention used in Ref. [1].

5. A Grassmann parity of an operator A is denoted by $|A| \in \{0, 1\}$.

1.1 Conventions for the bc ghost system

The reparametrization ghosts $b(z)$ and $c(z)$ are Grassmann odd. Their holomorphic conformal weights and ghost numbers are

field	h	ghost number
b	2	-1
c	-1	$+1$

The bc OPE is

$$b(z)c(w) \sim \frac{1}{z-w}. \tag{1.8}$$

The bc stress tensor is

$$T_{bc} := -2b\partial c - (\partial b)c, \tag{1.9}$$

and the central charge of the bc system is

$$c_{bc} = -26. \tag{1.10}$$

On the complex plane, a nonzero bc correlator contains three more c insertions than b insertions. We define the constant K by the normalization used in Ref. [1]:

$$\langle c(z_1)c(z_2)c(z_3)\rangle_{bc} := -K(z_1-z_2)(z_2-z_3)(z_1-z_3). \tag{1.11}$$

Here the subscript bc means that the expectation value is taken only in the bc sector.

1.2 Conventions for the $\beta\gamma$ system

The superconformal ghosts $\beta(z)$ and $\gamma(z)$ are Grassmann even and have conformal weights $3/2$ and $-1/2$, respectively. We bosonize them in terms of Grassmann-odd fields ξ, η and the Grassmann-even chiral scalar ϕ. The symbol $e^{q\phi}$ denotes the exponential of ϕ together with its cocycle factor, namely the z-independent factor that supplies the Grassmann parity required by the bosonization. For integer q, we define

$$\left|e^{q\phi}\right| := q \pmod 2, \qquad q \in \mathbb{Z}. \tag{1.12}$$

Thus $e^{q\phi}$ is Grassmann odd when q is odd. For half-integer q, a Grassmann parity is assigned only to a complete Ramond operator containing $e^{q\phi}$ and a spin field, rather than to $e^{q\phi}$ by itself. The bosonization convention is

$$\beta := \partial\xi\, e^{-\phi}, \tag{1.13}$$
$$\gamma := \eta\, e^{\phi}. \tag{1.14}$$

Equation (1.12) makes both $e^{-\phi}$ and e^{ϕ} Grassmann odd. Consequently, each right-hand side in Eqs. (1.13) and (1.14) is Grassmann even. The displayed factor order is part of the convention. The elementary operator products are

$$\xi(z)\eta(w) \sim \frac{1}{z-w}, \tag{1.15}$$
$$\partial\phi(z)\partial\phi(w) \sim -\frac{1}{(z-w)^2}, \tag{1.16}$$
$$e^{q_1\phi}(z)e^{q_2\phi}(w) \sim (z-w)^{-q_1 q_2} e^{(q_1+q_2)\phi}(w). \tag{1.17}$$

Differentiation of Eq. (1.15) gives

$$\begin{aligned}\partial\xi(z)\eta(w) &\sim -\frac{1}{(z-w)^2},\\ \eta(z)\partial\xi(w) &\sim \frac{1}{(z-w)^2}.\end{aligned}$$

Moving the odd exponential through the odd η or $\partial\xi$ therefore gives

$$\begin{aligned}\beta(z)\gamma(w) &= \partial\xi(z)e^{-\phi}(z)\eta(w)e^{\phi}(w)\\ &= -\partial\xi(z)\eta(w)e^{-\phi}(z)e^{\phi}(w) \sim \frac{1}{z-w},\\ \gamma(z)\beta(w) &= \eta(z)e^{\phi}(z)\partial\xi(w)e^{-\phi}(w)\\ &= -\eta(z)\partial\xi(w)e^{\phi}(z)e^{-\phi}(w) \sim -\frac{1}{z-w}.\end{aligned}$$

Thus

$$\begin{aligned}\beta(z)\gamma(w) &\sim \frac{1}{z-w},\\ \gamma(z)\beta(w) &\sim -\frac{1}{z-w}.\end{aligned} \tag{1.18}$$

The bosonized stress tensors are

$$T_{\xi\eta} := -\eta\partial\xi, \tag{1.19}$$
$$T_{\phi} := -\frac{1}{2}(\partial\phi)^2 - \partial^2\phi, \tag{1.20}$$
$$T_{\beta\gamma} := T_{\xi\eta} + T_{\phi}. \tag{1.21}$$

In the unbosonized variables, the same stress tensor is

$$T_{\beta\gamma} = \frac{3}{2}\beta\partial\gamma + \frac{1}{2}\gamma\partial\beta. \tag{1.22}$$

Its central charge is

$$c_{\beta\gamma} = 11. \tag{1.23}$$

The ghost number and picture number used below are additive charges defined on the elementary bosonized fields by the following table.

field	β	γ	ξ	η	$e^{q\phi}$
ghost number	-1	$+1$	-1	$+1$	0
picture number	0	0	$+1$	-1	q
Grassmann parity	0	0	1	1	$q \bmod 2$

The last entry in the Grassmann-parity row applies to $q \in \mathbb{Z}$. The conformal weight of the exponential, for $q \in \frac{1}{2}\mathbb{Z}$, is

$$h\big[e^{q\phi}\big] = -\frac{1}{2}q(q+2). \tag{1.24}$$

The mode expansions of the weight-zero field ξ and the weight-one field η define

$$\xi_0 := \oint \frac{\mathrm{d}z}{2\pi \mathrm{i} z}\,\xi(z), \qquad \eta_0 := \oint \frac{\mathrm{d}z}{2\pi \mathrm{i}}\,\eta(z). \tag{1.25}$$

The large Hilbert space is the state space on which ξ_0 acts without the restriction in Eq. (1.26). The small Hilbert space is the subspace of states $|\Psi\rangle$ satisfying

$$\eta_0|\Psi\rangle = 0. \tag{1.26}$$

1.3 Conventions for the BRST and PCO operators

In this subsection, T_{m} denotes the stress tensor of the matter CFT and T_F denotes the corresponding matter supercurrent. Their free-field expressions for the ten-dimensional matter CFT are given in Eqs. (1.48) and (1.49) below. The holomorphic BRST current is defined by

$$j_{\mathrm{B}} := c\big(T_{\mathrm{m}} + T_{\xi\eta} + T_\phi\big) + bc\,\partial c + \gamma T_F - \frac{1}{4}\gamma^2 b. \tag{1.27}$$

In bosonized variables, the composite field γ^2 is

$$\gamma^2 = \eta\,\partial\eta\,e^{2\phi}. \tag{1.28}$$

The BRST charge is the contour integral

$$Q_{\mathrm{B}} := \oint \frac{\mathrm{d}z}{2\pi\mathrm{i}}\, j_{\mathrm{B}}(z), \tag{1.29}$$

where the contour is counterclockwise around the operator on which Q_{B} acts. The anti-holomorphic charge $\widetilde{Q}_{\mathrm{B}}$ is defined by the tilded copy of Eqs. (1.27)–(1.29); the closed-string BRST charge is

$$Q_{\mathrm{closed}} := Q_{\mathrm{B}} + \widetilde{Q}_{\mathrm{B}}. \tag{1.30}$$

For two Grassmann-odd operators A and B, write $\{A, B\} := AB + BA$. The holomorphic picture-changing operator (PCO) is defined in the large Hilbert space by

$$\mathcal{X}(z) := 2\{Q_{\mathrm{B}}, \xi(z)\} \tag{1.31}$$

$$= 2c\,\partial\xi + 2e^{\phi}T_F - \frac{1}{2}\left[(\partial\eta)e^{2\phi}b + \partial\big(\eta e^{2\phi}b\big)\right](z). \tag{1.32}$$

The factor of two in Eq. (1.31) is part of the convention adopted from Ref. [1]. The PCO $\mathcal{X}$ has conformal weight zero, ghost number zero, picture number $+1$, and is BRST closed. The anti-holomorphic PCO is

$$\widetilde{\mathcal{X}}(\bar{z}) := 2\{\widetilde{Q}_{\mathrm{B}}, \widetilde{\xi}(\bar{z})\}. \tag{1.33}$$

For an oriented worldsheet of genus $g \in \mathbb{Z}_{\geq 0}$ with $b \in \mathbb{Z}_{\geq 0}$ boundary components, define P_{tot} to be the sum of the picture numbers of all holomorphic and anti-holomorphic insertions, including PCOs. A nonzero correlator requires

$$P_{\mathrm{tot}} = 2(2g + b - 2). \tag{1.34}$$

For a closed worldsheet, $b = 0$, this condition can be imposed separately as $P_{\mathrm{L}} = P_{\mathrm{R}} = 2g - 2$. Equation (1.34) explains why PCO insertions are required when the picture numbers of the vertex operators do not already have the required sum.

1.4 Conventions for the free boson

Let $X^M(z, \bar{z})$ be the ten embedding coordinates. We have the following OPEs:

$$X^M(z, \bar{z})X^N(w, \bar{w}) \sim -\frac{\eta^{MN}}{2}\log|z - w|^2, \tag{1.35}$$

$$\partial X^M(z)\partial X^N(w) \sim -\frac{\eta^{MN}}{2(z - w)^2}. \tag{1.36}$$

The holomorphic stress tensor of the ten bosons is defined by

$$T_X(z) := -\eta_{MN}\partial X^M \partial X^N(z). \tag{1.37}$$

With Eqs. (1.36) and (1.37), each X^M has holomorphic conformal weight zero, each ∂X^M has holomorphic conformal weight one, and the ten-boson central charge is

$$c_X = 10. \tag{1.38}$$

For a target-space momentum $k_M \in \mathbb{R}^{1,9}$, define

$$k \cdot X := k_M X^M, \tag{1.39}$$

$$k^2 := \eta^{MN} k_M k_N, \tag{1.40}$$

$$V_k(z,\bar{z}) := \exp\bigl(\mathrm{i}k \cdot X(z,\bar{z})\bigr). \tag{1.41}$$

The holomorphic and anti-holomorphic conformal weights of V_k are

$$(h_k, \bar{h}_k) = \left(\frac{k^2}{4}, \frac{k^2}{4}\right). \tag{1.42}$$

Two useful consequences of Eq. (1.35) are

$$\partial X^M(z) V_k(w,\bar{w}) \sim -\frac{\mathrm{i}k^M}{2(z-w)} V_k(w,\bar{w}), \tag{1.43}$$

$$V_k(z,\bar{z}) V_p(w,\bar{w}) \sim |z-w|^{k\cdot p} \exp\bigl(\mathrm{i}(k+p)\cdot X(w,\bar{w})\bigr), \tag{1.44}$$

where $k \cdot p := \eta^{MN} k_M p_N$.

1.5 Conventions for the free fermion

Let $\psi^M(z)$ and $\widetilde{\psi}^M(\bar{z})$ be the holomorphic and anti-holomorphic components of the ten real worldsheet fermions. Their normalization is

$$\psi^M(z)\psi^N(w) \sim -\frac{\eta^{MN}}{2(z-w)}. \tag{1.45}$$

The holomorphic fermion stress tensor is

$$T_\psi(z) := \eta_{MN}\psi^M \partial \psi^N(z). \tag{1.46}$$

Thus ψ^M has holomorphic conformal weight 1/2, and the ten-fermion central charge is

$$c_\psi = 5. \tag{1.47}$$

The total matter stress tensor and supercurrent are

$$T_{\mathrm{m}} := T_X + T_\psi = -\eta_{MN}\partial X^M \partial X^N + \eta_{MN}\psi^M \partial \psi^N, \tag{1.48}$$
$$T_F := -\eta_{MN}\psi^M \partial X^N. \tag{1.49}$$

We choose the convention in Eq. (1.49) to agree with Eq. (2.3) of Ref. [2].

We next fix the gamma matrices and the Ramond spin fields. Spinor indices α, β, γ take values in $\{1, \ldots, 16\}$. We use symmetric chiral gamma-matrix blocks

$$(\Gamma^M)_{\alpha\beta} = (\Gamma^M)_{\beta\alpha}, \qquad (\Gamma^M)^{\alpha\beta} = (\Gamma^M)^{\beta\alpha}. \tag{1.50}$$

Their normalization is defined by

$$(\Gamma^M)_{\alpha\beta}(\Gamma^N)^{\beta\gamma} + (\Gamma^N)_{\alpha\beta}(\Gamma^M)^{\beta\gamma} = 2\eta^{MN}\delta_\alpha{}^\gamma. \tag{1.51}$$

Equivalently, the 32×32 matrices

$$\mathbf{\Gamma}^M := \begin{pmatrix} 0 & (\Gamma^M)^{\alpha\beta} \\ (\Gamma^M)_{\alpha\beta} & 0 \end{pmatrix} \tag{1.52}$$

satisfy

$$\{\mathbf{\Gamma}^M, \mathbf{\Gamma}^N\} = 2\eta^{MN}\mathbf{1}_{32}. \tag{1.53}$$

For $p \geq 1$, the antisymmetrized product is

$$\mathbf{\Gamma}^{M_1\cdots M_p} := \mathbf{\Gamma}^{[M_1} \cdots \mathbf{\Gamma}^{M_p]}, \tag{1.54}$$

where the brackets include the factor $1/p!$. A symbol such as $(\Gamma^{M_1\cdots M_p})_\alpha{}^\beta$ denotes the indicated chiral block of Eq. (1.54). We choose the target-space orientation and positive-chirality block by

$$\epsilon^{01\cdots 9} = +1, \qquad (\Gamma^{01\cdots 9})_\alpha{}^\beta = \delta_\alpha{}^\beta. \tag{1.55}$$

Equations (1.50) and (1.55) are the ten-dimensional conventions of Ref. [3].

Let $S_\alpha(z)$ and $\widetilde{S}_\alpha(\bar{z})$ denote the left- and right-moving positive-chirality Ramond spin fields. Type IIB means that these two fields have the same ten-dimensional chirality. Each has conformal weight 5/8 in its chiral sector. The symbols S^α and $\widetilde{S}^\alpha$ denote the spin fields that occur in the $-3/2$ picture; the placement of the spinor index is part of this convention and is fixed by the operator products below, rather than by an unstated index-raising rule.

The Ramond operator products below use the scalar ϕ and the cocycle-dressed exponentials defined in Sec. 1.2. We normalize the spin fields by the singular terms

$$e^{-\phi}\psi^M(z)\; e^{-\phi/2}S_\alpha(w) \sim \frac{\mathrm{i}}{2(z-w)}(\Gamma^M)_{\alpha\beta}e^{-3\phi/2}S^\beta(w), \tag{1.56}$$

$$e^{-3\phi/2}S^\alpha(z)\; e^{-\phi/2}S_\beta(w) \sim \frac{\delta^\alpha{}_\beta}{(z-w)^2}e^{-2\phi}(w), \tag{1.57}$$

$$e^{-\phi/2}S_\alpha(z)\; e^{-\phi/2}S_\beta(w) \sim \frac{\mathrm{i}}{z-w}(\Gamma^M)_{\alpha\beta}e^{-\phi}\psi_M(w). \tag{1.58}$$

The branch choices, phases, and factors in Eqs. (1.56)–(1.58) are those of Ref. [3]. The anti-holomorphic spin fields obey the same equations with every field and coordinate replaced by its anti-holomorphic counterpart. This identical left-right form is another place where the type IIB chirality choice enters.

References

[1] Jyotirmoy Barman, Rishabh Kaushik, Raghu Mahajan, Chitraang Murdia, and Ashoke Sen, "Instanton-Induced Closed-String Amplitudes in Minimal Superstring Theory at Subleading Order," (2026), arXiv:2606.06596 [hep-th]

[2] Ashoke Sen, "Normalization of type IIB D-instanton amplitudes," JHEP **12**, 146 (2021), arXiv:2104.11109 [hep-th]

[3] Sergei Alexandrov, Ashoke Sen, and Bogdan Stefański, "Euclidean D-branes in Type IIB string theory on Calabi-Yau threefolds," JHEP **12**, 044 (2021), arXiv:2110.06949 [hep-th]